Microeconomics

Control Questions — Full Question Pool

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1 Chapter 1 — Recap and Basics

1 Stock and Flow Quantities *Multiple Choice*

Which of the following is a *flow* quantity?

(A) The total national debt of Germany.

(B) The oil reserves in a storage tank.

(C) ✓ The gross domestic product (GDP) per year.

(D) The number of cars registered in Bremen.

Solution

GDP per year is a flow quantity: it measures how much value is *added* over a period of time (here: one year). National debt, oil reserves, and the number of registered cars are all stock quantities — they describe how much of something exists at a point in time.

2 Price Elasticity of Demand *Multiple Choice*

The price elasticity of demand η for a good is measured at $\eta = -0.3$. Which statement is correct?

(A) Demand is elastic: a 1 % price rise causes a 3 % quantity drop.

(B) ✓ Demand is inelastic: a 1 % price rise causes only a 0.3 % quantity drop.

(C) Demand is unit elastic.

(D) The demand curve is horizontal.

Solution

$|\eta| = 0.3 < 1$, so demand is **inelastic**. The quantity demanded barely responds to the price change. Energy and petrol are classic examples of goods with inelastic demand.

3 Rule of 72 *Multiple Choice*

An economy grows at 4 % per year. Approximately how many years does it take to double in size, according to the Rule of 72?

(A) 4 years.

(B) 9 years.

(C) ✓ 18 years.

(D) 72 years.

Solution

The Rule of 72 states $\Delta t \approx 72/p$. At $p = 4\,\%$: $\Delta t \approx 72/4 = 18$ years.

4 Giffen Goods *True / False*

A Giffen good is a good for which demand *rises* when its price rises.

☑ **True** ☐ False

Solution

True. Giffen goods are an exception to the law of demand. Because they are strongly inferior goods consuming a large share of the consumer's budget, the income effect of a price rise dominates the substitution effect, so consumers actually buy *more*. Rice in subsistence economies is the classic textbook example.

5 Marginal vs. Average Values *True / False*

If the marginal value of producing one additional unit is *below* the current average, the average must be falling.

☑ **True** ☐ False

Solution

True. This is a purely mathematical relationship: whenever the next value added (marginal) is below the running average, it pulls the average down. Think of a class average: one very low new grade always lowers the mean.

6 Discounting and the Interest Rate *Open Question*

Explain in your own words why the interest rate can be interpreted as the *price of the future*. What happens to your valuation of a future payoff when the interest rate rises?

Solution

The interest rate reflects the opportunity cost of holding money in the present rather than investing it. A payment received in the future is worth less in today's terms the higher the interest rate is, because money invested now would grow quickly — the present value $K_0 = K_t/(1+r)^t$ falls as r rises. Conversely, at near-zero interest rates, the future is valued almost as highly as the present (the discount factor $1/(1+r)^t \approx 1$). High interest rates therefore encourage present consumption and discourage long-term investment.

2 Chapter 2 — The Consumer

7 Gossen's Second Law *Multiple Choice*

A utility-maximising consumer has spent her entire budget. Gossen's Second Law tells us that at the optimum:

(A) She buys only the cheapest good.

(B) ✓ The marginal utility per euro is equal across all goods purchased.

(C) The marginal utility of each good is zero.

(D) Total utility equals the budget divided by the price level.

Solution

Gossen's Second Law states that utility is maximised when the last euro spent on each good yields the same marginal utility. If this were not the case, the consumer could increase total utility by shifting spending from the lower-MU good to the higher-MU good.

8 Income and Substitution Effect *Multiple Choice*

The price of good X rises. For a *normal* good, the income effect and the substitution effect:

(A) Both lead to *more* consumption of X.

(B) Work in opposite directions; the income effect dominates.

(C) ✓ Both lead to *less* consumption of X.

(D) Work in opposite directions; the substitution effect dominates.

Solution

For a normal good, both effects reinforce each other. The **substitution effect** makes X relatively more expensive, so the consumer switches to alternatives. The **income effect** makes the consumer feel poorer, and since X is a normal good, less of it is consumed. Both effects reduce the quantity demanded.

9 Inferior and Superior Goods (Engel Curves) *Multiple Choice*

As a consumer's income rises, she buys *less* cheap instant noodles and *more* restaurant meals. This means:

(A) Both goods are normal goods.

(B) ✓ Instant noodles are an inferior good; restaurant meals are a superior (luxury) good.

(C) Both goods are inferior goods.

(D) Instant noodles are a Giffen good.

Solution

A good is **inferior** if consumption falls as income rises. A good is **superior** (or luxury) if consumption rises faster than income. Instant noodles are inferior here; restaurant meals are superior. Their Engel curves slope in opposite directions.

10 Cobb-Douglas Utility: $\alpha + \beta = 1$ *True / False*

In the Cobb-Douglas utility function $U = X^{\alpha} \cdot Y^{\beta}$ with $\alpha + \beta = 1$, the budget shares spent on X and Y are constant at α and β respectively, regardless of income or prices.

☑ **True** ☐ False

Solution

True. This is a standard result for Cobb-Douglas preferences. Optimising the budget gives $p_X \cdot x = \alpha \cdot B$ and $p_Y \cdot y = \beta \cdot B$, so the fraction of the budget spent on each good depends only on the exponents α and β, not on prices or income. Engel curves are linear and pass through the origin.

11 Hicks Decomposition *True / False*

The Hicks decomposition isolates the substitution effect by asking: how would the consumer re-allocate spending if relative prices changed but *real income* (utility) stayed the same?

☑ **True** ☐ False

Solution

True. In the Hicks (compensated) approach, we shift the new budget line outward until it is tangent to the *old* indifference curve. The movement along that curve to the new tangency point is the pure substitution effect; the remaining change (between the two budget lines) is the income effect.

12 Perfect Complements vs. Perfect Substitutes *Open Question*

Compare perfect complements and perfect substitutes in terms of their indifference-curve shapes, their marginal rates of substitution, and what happens to the income and substitution effects when the price of one good rises.

Solution

Perfect complements (e.g. left and right shoes) have L-shaped indifference curves. The MRS is zero except at the kink. A price rise of one good causes a pure *income effect*: the consumer becomes poorer and buys less of both goods in fixed proportion. There is no substitution effect.

Perfect substitutes (e.g. two identical shops selling fish and chips) have linear indifference curves. The MRS is constant ($\sigma = \infty$ in the limit). A price rise of one good causes a pure *substitution effect*: the consumer switches entirely to the cheaper substitute. There is no income effect (real utility is unchanged).

The normal, in-between case (Cobb-Douglas) exhibits both effects partially.

3 Chapter 3 — Demand and Supply

13 The Supply Curve under Competition *Multiple Choice*

Under perfect competition in the short run, the firm's supply curve is:

(A) The average total cost curve.

(B) ✓ The marginal cost curve (above average variable cost).

(C) Horizontal, because all firms are price takers.

(D) Downward-sloping due to economies of scale.

Solution

A competitive firm sets output where $P = MC$. The supply curve is therefore the **marginal cost curve**. The firm only supplies if price covers variable costs, so the relevant portion is above average variable cost.

14 Elasticity of Supply — Short vs. Long Run *Multiple Choice*

Why is the price elasticity of supply typically *lower* in the short run than in the long run?

(A) ✓ In the short run, capacity is fixed; firms cannot quickly expand production in response to a price rise.

(B) In the short run, demand is more elastic, which reduces supply elasticity.

(C) More firms operate in the short run, reducing each firm's ability to adjust.

(D) Fixed costs do not exist in the long run.

Solution

Short-run supply is inelastic because physical capacity cannot be changed quickly. Over the long run, new capacity can be built and new firms can enter, making supply much more responsive to price changes.

15 Edgeworth Box and the Trading Curve *Multiple Choice*

In an Edgeworth box with two goods and two traders, the *contract curve* (trading curve) is:

(A) The diagonal of the box.

(B) ✓ The locus of all points where both traders' indifference curves are tangent — any move away harms at least one party.

(C) The locus of points where both traders are at the same utility level.

(D) The budget line shared by both traders.

Solution

The **contract curve** connects all Pareto-efficient allocations: points where the two indifference curves are tangent (equal MRS). Any point on the curve is stable because moving away would make at least one party worse off. Any point off the curve is inefficient: both parties could gain by trading.

16 The Pork Cycle *True / False*

The pork cycle arises because supply reacts to current prices with a time lag, generating cyclical price and quantity dynamics.

☑ **True** ☐ False

Solution

True. High pork prices induce farmers to raise more pigs; supply increases only when the animals mature. Excess supply then depresses prices, triggering herd reductions — and so on. The same cobweb mechanism occurs in shipping, construction, and commodity markets.

17 Merit Order in Energy Markets *True / False*

In the energy merit order, the price of electricity is determined by the marginal cost of the *most expensive* technology needed to meet demand.

☑ **True** ☐ False

Solution

True. Technologies are stacked from cheapest to most expensive (renewables, coal, gas). The last — most expensive — technology dispatched sets the market price. All cheaper technologies therefore earn rents (producer surplus). When renewable supply is very large and covers all demand, the price can fall to near zero or below.

18 The Pork Cycle *Open Question*

Describe the pork cycle. Why does it arise, and what broader economic lesson does it illustrate?

Solution

When pork prices are high, many farmers invest in raising more pigs. Because pigs take time to grow, supply only reaches the market with a lag of roughly 1–2 years. The surplus then depresses prices, causing farmers to reduce herds. With less supply later, prices rise again — and the cycle repeats.

The pork cycle illustrates that **markets with production lags can generate endogenous oscillations** (cobweb dynamics) even without external shocks. The same pattern underlies boom-bust cycles in shipping, construction, and commodity markets.

4 Chapter 4 — Equilibrium and Welfare

19 Consumer and Producer Surplus *Multiple Choice*

In a competitive equilibrium, the consumer surplus is:

(A) The area below the supply curve and above the equilibrium price.

(B) ✓ The area below the demand curve and above the equilibrium price.

(C) Equal to the producer surplus.

(D) The total revenue of all producers.

Solution

Consumer surplus is the difference between what buyers are *willing to pay* (their reservation prices, read off the demand curve) and what they *actually pay* (the equilibrium price). Graphically it is the triangle below DD and above p^*.

20 Welfare and Pareto Efficiency *Multiple Choice*

The outcome of a perfectly competitive market is Pareto optimal. This means:

(A) Everyone has the same income.

(B) ✓ No reallocation can make one party better off without making another worse off.

(C) Total output is maximised regardless of how it is distributed.

(D) The government has intervened to correct market failures.

Solution

Pareto efficiency means we have exhausted all mutually beneficial trades. In a competitive equilibrium, all surplus-generating trades are executed, and the sum of consumer and producer surplus is maximised. It says nothing about the *fairness* of the distribution.

21 Elasticity and Who Captures the Surplus *Multiple Choice*

Demand is highly *inelastic* and supply is *elastic*. Which party captures the larger share of the total welfare gain?

(A) ✓ Consumers, because they are willing to pay far more than the equilibrium price.

(B) Producers, because they have market power.

(C) The surplus is split equally.

(D) No surplus exists if supply is elastic.

Solution

When demand is inelastic, many consumers would have paid a much higher price, so the area below DD and above p^* (consumer surplus) is large relative to the narrow triangle of producer surplus. Examples: energy, insulin.

22 Reservation Price *True / False*

A consumer's *reservation price* for a good is the maximum price she would be willing to pay for it; if the market price is below this, she earns a positive consumer surplus.

☑ **True** ☐ False

Solution

True. Consumer surplus = reservation price − market price. If the market price equals or exceeds the reservation price, the consumer does not buy (zero or negative surplus). The competitive equilibrium aggregates all such individual differences into total consumer surplus.

23 Pareto and Kaldor-Hicks *True / False*

A policy change that makes some people better off and others worse off *cannot* be justified under the Pareto criterion, but it *can* be justified under the Kaldor-Hicks criterion if the winners could in principle compensate the losers.

☑ **True** ☐ False

Solution

True. The strict Pareto criterion requires that nobody be made worse off. Kaldor-Hicks relaxes this: a change is desirable if the gains to winners are large enough to *potentially* compensate losers — even if no actual compensation takes place.

24 Interpreting Consumer and Producer Surplus *Open Question*

Using the demand-supply diagram, explain what consumer surplus and producer surplus represent. Under what circumstances is one much larger than the other, and what real-world examples

illustrate this?

Solution

Consumer surplus (CS) is the sum of each buyer's reservation price minus the equilibrium price: the benefit consumers obtain beyond what they pay. Graphically it is the area below the demand curve and above p^*.

Producer surplus (PS) is the amount above the minimum price sellers would accept (their marginal costs): the area above the supply curve and below p^*. If there are no fixed costs, it equals profit.

CS is large relative to **PS** when demand is inelastic (e.g. petrol, life-saving medicines): buyers have few substitutes and would pay much more than the market price.

PS dominates when supply is inelastic (e.g. old masters paintings, prime real estate): the quantity is naturally limited, so sellers capture rents.

5 Chapter 5 — Market Disruptions

25 Ad Valorem vs. Per-Unit Tax *Multiple Choice*

An ad valorem tax on a good differs from a per-unit tax in that:

(A) It is paid by consumers, not producers.

(B) ✓ It is proportional to the value of the good, causing the supply curve to *rotate* rather than shift in parallel.

(C) It always generates more government revenue.

(D) It has no effect on consumer surplus.

Solution

A **per-unit tax** adds a fixed amount T to each unit, shifting the supply curve upward by T in parallel. An **ad valorem tax** levies a percentage of price, so the mark-up is larger at high prices, rotating the supply curve. Both can in principle produce the same equilibrium outcome for the right parameter choice.

26 Tax Burden and Elasticity *Multiple Choice*

A unit tax is imposed on a market where demand is highly inelastic and supply is elastic. Who bears the greater share of the tax burden?

(A) Producers, because they choose the price.

(B) ✓ Consumers, because they cannot easily reduce their quantity.

(C) The burden is always split 50:50 regardless of elasticities.

(D) Nobody; the deadweight loss absorbs the entire tax.

Solution

The **less elastic** party bears more of the burden. With inelastic demand, consumers have few substitutes and cannot avoid the tax, so most of the price increase passes to them. This is why taxes on energy, tobacco, and alcohol are fiscally attractive: they generate stable revenue with limited market distortion.

27 Rigid Quantity Restriction *Multiple Choice*

A government caps the quantity of a good that may be sold to $x_2 < x^*$ (the competitive quantity). Compared to the unregulated equilibrium:

(A) The price falls and welfare increases.

(B) Only consumer surplus falls; producer surplus is unchanged.

(C) ✓ Both consumer and producer surplus change; total welfare falls and a deadweight loss arises.

(D) The quantity restriction has no effect if demand is elastic.

Solution

A quota below x^* raises the market price above p^*. Consumers who are excluded lose surplus. The net effect on producer surplus depends on the elasticities: some producers gain from the higher price, but others are forced out of the market. Total welfare falls because mutually beneficial trades are prevented — this is the deadweight loss triangle.

28 Supply Shocks Are Always Passed On in Full *True / False*

A cost shock (e.g. a sudden rise in raw-material prices) that shifts the supply curve upward is always passed on entirely to consumers in the form of a higher market price.

☐ True ☑ **False**

Solution

False. How much of a cost shock is passed on depends on the elasticities of supply and demand. The less elastic party bears more of the burden. In the symmetric case, the burden is shared equally. A cost shock will generally *never* be passed on 100 % if demand has any negative slope — producers absorb part of it through lower margins.

29 Deadweight Loss from a Tax *True / False*

A tax always creates a deadweight loss — a portion of welfare that is completely destroyed rather than transferred to the government.

☑ **True** ☐ False

Solution

True. A tax raises the price and reduces the quantity traded below the efficient level. The transactions that no longer take place would have generated surplus for both buyer and seller, but that potential gain is simply lost. Government revenue (GR) compensates partly, but the triangle of deadweight loss DWL is not captured by anyone.

30 Welfare Effects of a Supply Shock *Open Question*

A per-unit cost increase shifts the supply curve upward. Using a supply-demand diagram, describe how this affects price, quantity, consumer surplus, producer surplus, and total welfare. Explain the role of elasticities.

Solution

The supply curve shifts up by the size of the cost increase. In the new equilibrium: price rises from p_1 to p_2 and quantity falls from x_1 to x_2.

Consumer surplus falls: buyers pay more and fewer units are traded. **Producer surplus** falls too: although price rose, their costs rose by more, and fewer units are sold. **Total welfare** falls: the new combined surplus is smaller than before, with a deadweight loss triangle.

Elasticities determine the split: the less elastic party bears more of the burden. With inelastic demand (e.g. petrol), consumers bear most of the cost increase. With inelastic supply, producers bear more.

6 Chapter 6 — Monopolies: The Cournot Model

31 Cournot Point *Multiple Choice*

A profit-maximising monopolist sets output where:

(A) Price equals average cost.

(B) Price equals marginal cost.

(C) ✓ Marginal revenue equals marginal cost ($MR = MC$).

(D) Total revenue is maximised.

Solution

The monopolist maximises $\Pi(x) = R(x) - C(x)$ by setting $dR/dx = dC/dx$, i.e. $MR = MC$. This is the **Cournot point**. Price equals marginal cost only under perfect competition. Total revenue is maximised at unit elasticity, not at the profit maximum.

32 Monopoly Output vs. Competitive Output *Multiple Choice*

With a linear demand curve and constant marginal costs, a monopolist produces approximately:

(A) The same quantity as perfect competition.

(B) Twice the competitive quantity.

(C) ✓ Half the competitive quantity.

(D) Three-quarters of the competitive quantity.

Solution

For a linear demand curve $P = a - bQ$ and constant $MC = c$, the competitive output is $Q_C = (a - c)/b$ and the monopoly output is $Q_M = (a - c)/(2b) = Q_C/2$. The monopoly price is roughly twice the competitive price.

33 First-Degree Price Discrimination *Multiple Choice*

A firm practising first-degree (perfect) price discrimination charges each buyer their reservation price. The result is:

(A) No transactions take place because prices are too high.

(B) Consumer surplus is maximised at the expense of producer surplus.

(C) ✓ Consumer surplus is zero and all surplus is captured as producer surplus; the quantity sold equals the competitive level.

(D) A larger deadweight loss than under standard monopoly pricing.

Solution

With perfect price discrimination, the monopolist extracts the full value from every buyer. Consumer surplus falls to zero, but every consumer who values the good above marginal cost still buys — so quantity equals the competitive level and there is no deadweight loss. In practice this requires perfect information about willingness to pay and no resale between buyers.

34 Monopoly and Welfare *True / False*

Compared to perfect competition, a monopoly results in a higher price, a lower quantity traded, and a net loss of social welfare.

☑ **True** ☐ False

Solution

True. The monopolist restricts output to $Q_M < Q_C$ and charges $P_M > P_C$. Consumer surplus falls sharply, and the gap between the competitive and monopoly outcomes that is captured by neither party is the **deadweight welfare loss**. Producer surplus (profit) rises, but total welfare is lower than under competition.

35 Herfindahl-Hirschman Index (HHI) *True / False*

The Herfindahl-Hirschman Index $H = \sum_{i=1}^{N} a_i^2$ (where a_i is each firm's market share) reaches its maximum value of 1 when the market is a perfect monopoly.

☑ **True** ☐ False

Solution

True. If there is one firm with $a_1 = 1$ (100 %), then $H = 1^2 = 1$. In perfect competition with infinitely many tiny firms, $H \to 0$. Competition authorities use the HHI to assess market concentration when evaluating mergers.

36 Arguments For and Against Monopolies *Open Question*

Give one economic argument *in favour of* allowing a natural monopoly, and one argument *against* it. What does economic theory suggest as the best policy response?

Solution

In favour: Natural monopolies arise where fixed costs are so high that a single firm can supply the entire market more cheaply than two competing firms (e.g. railway networks, electricity grids). Duplicating the infrastructure would waste resources.

Against: An unregulated monopolist extracts rents, produces less than is socially optimal, and has little incentive to be efficient or innovate.

Policy: The theoretical best response (Jean Tirole) is *regulation* — capping the monopolist's rate of return or price — rather than nationalisation or break-up, since a change of ownership alone does not necessarily improve incentives or outcomes.

7 Chapter 8 — Market Imperfections

37 Adverse Selection *Multiple Choice*

Adverse selection in insurance markets arises because:

(A) Insurance companies invest premiums poorly.

(B) ✓ Those who most want insurance are those most likely to make a claim, driving up premiums and driving low-risk customers out.

(C) Insured individuals start behaving more recklessly after taking out a policy.

(D) Governments subsidise insurance companies.

Solution

Adverse selection (Akerlof, 1970) is a pre-contractual information problem: high-risk individuals have greater incentive to buy insurance than low-risk ones. As the insurer cannot distinguish between the two, it must charge a higher premium, which drives out more low-risk customers, raising the average risk further. In the extreme, markets can collapse entirely (the "market for lemons").

38 Pigouvian Tax *Multiple Choice*

A government introduces a Pigouvian tax equal to the marginal *external* cost of a polluting industry. The correct new equilibrium achieves:

(A) Zero production, since any pollution is unacceptable.

(B) Maximum government revenue.

(C) ✓ Production at the socially optimal level where price reflects full social costs.

(D) An outcome identical to the untaxed market.

Solution

The Pigouvian tax internalises the externality by raising private marginal cost to match social marginal cost. Firms reduce output to the social optimum, where price reflects the full cost — including environmental damage. The government also collects revenue that can compensate those harmed.

39 Tragedy of the Commons *Multiple Choice*

The tragedy of the commons occurs when:

(A) A good is excludable but non-rival, so firms under-produce it.

(B) ✓ A resource is non-excludable and rival, so individuals overuse it because private gain exceeds private cost, ignoring the communal cost.

(C) A monopolist restricts output of a public good.

(D) International trade destroys a domestic commons.

Solution

Common-pool resources are **rival** (my catch reduces yours) but **non-excludable** (you cannot stop me fishing). Each individual's private benefit from over-fishing exceeds their private cost, but the communal cost is ignored. Solutions include privatisation, regulation, or — as shown by Elinor Ostrom — community governance without either.

40 Moral Hazard *True / False*

Moral hazard is an example of a *post-contractual* information problem: once insured, a person may take more risks than they would without insurance.

☑ **True** ☐ False

Solution

True. Moral hazard arises *after* a contract is signed: the insured party's behaviour changes because they no longer bear the full consequences of their actions. By contrast, adverse selection is a *pre-contractual* problem (hidden type, not hidden action).

41 Coase Theorem *True / False*

The Coase Theorem states that, provided transaction costs are zero and property rights are clearly defined, parties will bargain to an efficient outcome regardless of who is initially assigned the

property right.

☑ **True** ☐ False

Solution

True. The key insight is that the *initial* assignment of property rights affects distribution but not efficiency (if bargaining is costless). In practice, transaction costs are never zero, property rights may be contested, and bargaining power is often unequal — which is why Elinor Ostrom's work on commons governance remains important.

42 Asymmetric Information and the Market for Lemons *Open Question*

Explain Akerlof's "market for lemons" argument. Why can asymmetric information cause a market to function poorly or even collapse? Give a real-world example beyond used cars.

Solution

Akerlof observed that in the used-car market, sellers know the true quality of their car but buyers do not. Buyers therefore offer only the *average* price for a random car. Owners of high-quality cars (*"peaches"*) find this average price too low and withdraw from the market. The average quality of cars on offer falls, so buyers lower their willingness to pay further — a downward spiral that can empty the market of good cars entirely.

Mechanisms: Adverse selection (bad drives out good) and the absence of credible quality signals. Solutions include warranties, certification, regulation, or repeated interaction.

Real-world example: Labour markets (workers know their own productivity better than employers), insurance (only the high-risk seek coverage), or credit markets (only risky borrowers may accept high interest rates).

8 Chapter 9 — The State

43 Price Ceiling *Multiple Choice*

A government sets a *price ceiling below* the competitive equilibrium price. The most likely result is:

(A) A surplus of goods as producers over-supply at the low price.

(B) ✓ Excess demand — more buyers want the good at the low price than sellers are willing to supply.

(C) No change, because markets always clear at the equilibrium price.

(D) An increase in total welfare.

Solution

A binding price ceiling is below p^*. At the lower price, demand exceeds supply (**excess demand**). Some buyers will not find the good, creating queues and potentially black markets. Historical examples include Emperor Diocletian's bread price ceiling, which contributed to shortages.

44 Laffer Curve *Multiple Choice*

The Laffer curve illustrates that:

(A) Tax revenue rises monotonically with the tax rate.

(B) Tax revenue falls monotonically with the tax rate.

(C) ✓ Tax revenue is zero at both a 0 % and 100 % tax rate, implying a revenue-maximising rate somewhere in between.

(D) A flat tax always maximises government revenue.

Solution

At 0 % tax, revenue is zero. At 100 %, all economic activity ceases and revenue is again zero. Tax revenue must therefore reach a maximum at some intermediate rate. The exact location of that maximum is empirically disputed and subject to income, substitution, and avoidance effects.

45 Ramsey Pricing Rule *Multiple Choice*

The Ramsey rule for optimal taxation states that, to minimise welfare loss, the government should:

(A) Tax all goods at the same percentage rate.

(B) ✓ Tax goods with lower price elasticity more heavily, since buyers cannot avoid the tax easily.

(C) Tax goods with higher price elasticity more heavily, to maximise revenue.

(D) Only tax goods that are luxury items.

Solution

The Ramsey rule says to tax more heavily where the distortion is smallest, i.e. where demand (or supply) is least elastic. Highly inelastic goods (energy, tobacco) generate revenue with little reduction in traded quantity — minimising deadweight loss. A lump-sum (per-head) tax is the extreme case: it cannot be avoided at all, so it causes no market distortion.

46 Who Pays a Tax — Producer or Consumer? *True / False*

Whether a unit tax is legally collected from the producer or the consumer makes no difference to the ultimate economic incidence of the tax.

☑ **True** ☐ False

Solution

True. The *statutory* incidence (who writes the cheque) differs from the *economic* incidence (who bears the cost). A tax on producers shifts the supply curve up; a tax on consumers shifts the demand curve down. Both lead to the same new equilibrium price and quantity and the same split of the burden between the two sides of the market.

47 Subsidies Always Increase Net Welfare *True / False*

Subsidising a market always increases total net social welfare.

☐ True ☑ **False**

Solution

False. A subsidy lowers the market price and increases quantity traded, raising welfare *in that market*. However, the subsidy must be funded — through taxes elsewhere — which creates an excess burden in other markets. Producers also lose the incentive to reduce costs. The total subsidy paid exceeds the welfare gain in the subsidised market, so there is a net social cost.

48 Price Ceilings: Why Do They Often Backfire? *Open Question*

Analyse the welfare effects of a binding price ceiling below the market equilibrium price. Explain what excess demand is, what happens to welfare, and why black markets may emerge.

Solution

A binding price ceiling $\bar{p} < p^*$ reduces the quantity supplied (some sellers exit) while increasing quantity demanded. The result is **excess demand**: a gap $F - G$ between those who want to buy and those willing to sell at $\bar{p}$.

Welfare effects: Consumer surplus is redistributed and partly destroyed — some buyers gain (those who can buy at the lower price) but many buyers are rationed. Producer surplus falls sharply. Total welfare is lower than at the competitive equilibrium; the lost triangle is a deadweight loss.

Black markets arise because buyers willing to pay above $\bar{p}$ and sellers willing to sell above $\bar{p}$ have an incentive to transact illegally. Enforcement is costly and may be ineffective, as shown historically by rent controls and wartime price controls.

9 Chapter 10 — Game Theory and Duopolies

49 Nash Equilibrium *Multiple Choice*

A Nash equilibrium in a game is a situation in which:

(A) Both players achieve their maximum possible payoff.

(B) ✓ No player can increase their own payoff by unilaterally changing their strategy.

(C) Both players cooperate to maximise total welfare.

(D) The dominant strategy of each player is to defect.

Solution

A **Nash equilibrium** is a strategy profile in which each player's choice is a best response to the others' choices. No individual has an incentive to deviate unilaterally. Note: it need not maximise total welfare — the Prisoner's Dilemma demonstrates this clearly.

50 Prisoner's Dilemma *Multiple Choice*

In the one-shot Prisoner's Dilemma, the Nash equilibrium is:

(A) Both players cooperate.

(B) ✓ Both players defect, even though mutual cooperation would be better for both.

(C) One player cooperates and the other defects.

(D) The players randomise equally between cooperate and defect.

Solution

Mutual defection is the Nash equilibrium: defecting is a dominant strategy (it yields a higher payoff regardless of the other player's choice). Yet mutual cooperation would be better for both — the dilemma. In repeated play, cooperation can be sustained by strategies such as *tit for tat*.

51 Stackelberg Leadership *Multiple Choice*

In a Stackelberg duopoly, the *follower* faces:

(A) The full market demand curve.

(B) A demand curve steeper than the leader's.

(C) ✓ A *residual* demand curve — the part of market demand not already served by the leader.

(D) No demand at all, since the leader captures the whole market.

Solution

The Stackelberg leader moves first and serves a portion of the market. The follower then acts as a de-facto monopolist over the *residual demand* — what remains. The leader anticipates this and chooses a large quantity to leave little residual demand for the follower, securing a first-mover advantage. Example: Amazon vs. Otto.de.

52 Battle of the Sexes: Two Nash Equilibria *True / False*

The Battle of the Sexes game has exactly two Nash equilibria in pure strategies.

☑ **True** ☐ False

Solution

True. In the Battle of the Sexes, Alice prefers (Concert, Concert) and Bob prefers (Football, Football), but both prefer to be together rather than apart. Both (Concert, Concert) and (Football, Football) are Nash equilibria in pure strategies — neither player wants to deviate unilaterally once they coordinate. There is also a mixed-strategy equilibrium.

53 Predatory Pricing *True / False*

Predatory pricing — setting a price so low that rivals cannot survive — creates long-term harm to consumers even if it appears to help them in the short run.

☑ **True** ☐ False

Solution

True. The predatory firm accepts short-run losses to drive out competitors, after which it raises prices to monopoly level, recouping losses at the consumers' expense. This is why predatory pricing is mostly illegal under competition law in the EU and the US.

54 Why Does the Prisoner's Dilemma Matter for Trade Policy? *Open Question*

Explain the logic of the Prisoner's Dilemma and apply it to the decision of two countries over whether to impose trade tariffs on each other.

Solution

In the Prisoner's Dilemma, mutual cooperation yields the best collective outcome, but each player has a private incentive to defect regardless of the other's action — so rational self-interest leads to a collectively inferior outcome.

Applied to trade: If both countries maintain free trade (cooperate), both gain from exchange. But if one country defects by imposing tariffs, it protects its domestic producers at the expense of its trading partner. Because both face this incentive, the Nash equilibrium is mutual tariffs — a trade war that leaves both countries worse off than free trade would.

Solution: International agreements (WTO, bilateral treaties) act as commitment devices that make sustained cooperation stable, mimicking the "tit for tat" outcome in a repeated game.

10 Chapter 11 — Money and Trade

55 Absolute Advantage *Multiple Choice*

Country A can produce both fish and fruit more efficiently than Country B. According to the principle of *absolute* advantage:

(A) Trade is only beneficial for Country A.

(B) No trade will take place because Country B has nothing to offer.

(C) ✓ Trade can still be mutually beneficial if each country specialises in the good it produces relatively more efficiently.

(D) Country B should impose tariffs to protect its industries.

Solution

Absolute advantage says who produces a good *cheaply in absolute terms*; comparative advantage (Ricardo) says who produces it *cheaply relative to other goods*. Even if Country A is better at producing everything, trade based on comparative advantage benefits both countries, as each specialises where its opportunity cost is lowest.

56 Comparative Advantage *Multiple Choice*

David Ricardo's principle of comparative advantage states that:

(A) Only countries with an absolute advantage should export goods.

(B) Specialisation is only beneficial if one country is better at producing all goods.

(C) ✓ Each country should specialise in the good for which its *opportunity cost* is lowest, regardless of absolute productivity levels.

(D) Free trade maximises welfare only when productivity is equal between trading partners.

Solution

Ricardo showed that **comparative advantage** — not absolute advantage — drives the gains from trade. A country with lower opportunity cost in producing a good should export that good, even if the other country is absolutely more productive in every sector. Both countries end up consuming more than they could in autarky.

57 Trade and the Production Possibility Frontier *Multiple Choice*

When two countries trade based on comparative advantage:

(A) The combined production possibility frontier contracts.

(B) One country gains but the other loses.

(C) ✓ The combined PPF expands, allowing both countries to consume beyond their individual production frontiers.

(D) Specialisation is complete for both countries.

Solution

Trade effectively *pools* the two production frontiers. The combined frontier lies outside both individual frontiers, allowing consumption combinations otherwise unattainable. Typically, the less productive country specialises fully, while the more productive country may still produce some of both goods.

58 Who Gains and Who Loses from Trade? *True / False*

Opening a country to trade always makes every single citizen of that country better off.

☐ True ☑ **False**

Solution

False. Free trade increases *aggregate* welfare, but the gains and losses are distributed unevenly. In the fish-fruit example, island fruit producers and continental fishermen are displaced. Compensation is possible in principle (Kaldor-Hicks), but does not automatically occur. Political resistance to trade often reflects these distributional effects.

59 Comparative Advantage: Small Country *True / False*

When a small island begins trading with a large continent, the impact on the continent's prices and trade structure is negligible, while the island undergoes complete specialisation.

☑ **True** ☐ False

Solution

True. A small country is a price-taker on the world market: its trade volumes are too small to move continental prices. The island will fully specialise in its comparative-advantage good (e.g. fish), while the continent's relative price of fish shifts only marginally.

60 Absolute vs. Comparative Advantage *Open Question*

Distinguish between absolute and comparative advantage. Use the island-continent fish-and-fruit example to explain why comparative advantage, not absolute advantage, determines the pattern of trade and specialisation.

Solution

Absolute advantage: one party produces a good using fewer inputs than another (e.g. the continent is better at *both* fish and fruit).

Comparative advantage: one party produces a good at a lower *opportunity cost* than another. Even if the continent is absolutely more efficient in both, the island may be relatively better at fishing than at growing fruit.

In the example, the island trades fish at 1:4 (one fish for four fruit) internally, but can get the same fish at 1:1 on the continent. Island fishers gain dramatically by exporting. Island fruit farmers are out-competed and move into fishing. The island specialises totally in fish.

Key insight (Ricardo): It is *relative* (opportunity) costs, not absolute productivity, that drive trade. Both parties consume more than in autarky: a genuine "gains from trade".

11 Chapter 12 — Protectionism

61 Effect of an Import Tariff *Multiple Choice*

A government imposes a tariff on imports. Compared to free trade, the tariff:

(A) Increases consumer surplus and decreases producer surplus.

(B) ✓ Decreases consumer surplus, increases producer surplus and government revenue, but creates a net deadweight loss.

(C) Raises the world price of the good.

(D) Has no effect on domestic producers.

Solution

A tariff raises the domestic price above the world price. Consumers pay more and buy less (**CS falls**). Domestic producers gain (**PS rises**). The government collects tariff revenue (**GR**). However, the total of CS + PS + GR is less than the original CS + PS under free trade: the two small triangles that are lost to nobody form the **deadweight loss**.

62 Quota vs. Tariff *Multiple Choice*

An import quota and a tariff can, in principle, produce the same equilibrium price and quantity. A key difference is:

(A) A quota never reduces consumer surplus; a tariff always does.

(B) ✓ Under a quota, the government does not automatically receive revenue; the benefit may accrue to importers as quota rents.

(C) A tariff affects domestic producers; a quota does not.

(D) Quotas are always more efficient than tariffs.

Solution

Both instruments restrict imports to the same level and produce the same market price. The difference is in **who captures the rents**: with a tariff, the government captures the revenue. With a quota, importers who receive licences keep the rents (the gap between world price and domestic price times the quota volume). If quotas are auctioned, the equivalence to tariffs is restored.

63 Free Trade and Welfare *Multiple Choice*

From the perspective of economic theory, free trade is:

(A) Always bad for small countries.

(B) Only beneficial if both trading partners are equally productive.

(C) ✓ Welfare-maximising in the aggregate, even though distributional effects create losers within each country.

(D) Only beneficial when there are no trade deficits.

Solution

Theory shows that free trade maximises aggregate global welfare. Trade restrictions (tariffs, quotas, subsidies) create deadweight losses. However, the gains and losses are unevenly distributed, which generates political pressure for protectionism. Compensation mechanisms (redistribution) can in principle make everyone better off, but this rarely happens in practice.

64 Effect of a Tariff on a Small Country *True / False*

A small country imposing an import tariff cannot influence the world price; it only raises the domestic price, so all welfare losses fall on domestic consumers and the deadweight loss is borne entirely domestically.

☑ **True** ☐ False

Solution

True. A small country is a price taker: its tariff does not shift world supply and hence does not affect the world price. The tariff drives a wedge between the world price and the domestic price. Domestic consumers lose, domestic producers gain, the government gains revenue, but the net effect is a domestic deadweight loss. A large country can theoretically use an "optimum tariff" to extract rents from foreign exporters.

65 Export Subsidy as a Trade Policy Tool *True / False*

An export subsidy makes domestic products cheaper for foreign buyers, increases domestic production, but reduces welfare in the subsidising country because the cost of the subsidy exceeds the welfare gain.

☑ **True** ☐ False

Solution

True. An export subsidy shifts the supply curve down (or equivalently, the export supply curve). More is exported at a lower price for foreign buyers. Domestic consumers face a higher domestic price (since producers prefer to export). The net welfare effect in the subsidising country is negative: the subsidy cost is borne by taxpayers and exceeds the gain in producer surplus. Trading partners typically retaliate or challenge such subsidies under WTO rules.

66 Why Do Countries Use Protectionist Policies Despite the Theoretical Case for Free Trade? *Open Question*

Outline the main theoretical arguments for free trade. Then explain why governments still resort to tariffs, quotas, and subsidies, using the concepts of distributional effects, political economy, and the Prisoner's Dilemma.

Solution

The case for free trade: Based on comparative advantage, trade expands the combined production frontier and raises aggregate welfare in all participating countries. Any tariff or quota creates a domestic deadweight loss, and retaliatory tariffs reduce welfare further for all.

Why protectionism persists:

1. **Distributional effects**: Trade creates winners (competitive exporters) and losers (import-competing industries). Losers are often geographically concentrated and politically organised; winners are diffuse. This asymmetry creates lobbying pressure.

2. **Political economy**: Governments face pressure from concentrated interest groups. Tariffs are visible, targeted transfers to specific industries. Subsidies serve similar functions.

3. **Prisoner's Dilemma**: Each country has a short-term incentive to defect from free trade, even though mutual free trade is the cooperative optimum. Without binding international agreements (WTO), the Nash equilibrium is mutual tariffs — a trade war.

Conclusion: International trade agreements, by making commitments credible, help countries overcome the Prisoner's Dilemma and sustain the cooperative (free-trade) equilibrium.